

Development Plan

Software Engineering

Team 6, Blockchainers

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Table 1: Revision History

Date	Developer(s)	Change
September 27	Nathan	Changed project decomposition and scheduling, changed GitHub Projects usage, changed coding standards, changed Appendix – Team Charter
September 27	Aidan	Added sections 5 to 7 of the development plan regarding Team communication plan, roles, and workflow.
September 27	Keira	Added section 13, the reflection, and part of the team charter to the report.
September 28	Robert	Added sections 1-4 and 9-10
September 28	Keira	Revised section 13

This document outlines the processes, tools, and practices that the team will use to plan, develop, and manage the [DAIIM](#) project. It describes the team’s approach to communication, collaboration, project scheduling, workflow, and software development, as well as the technologies and coding standards that will guide implementation. The plan also identifies the project’s proof-of-concept strategy and describes how generative AI and other development tools will be incorporated into the development process.

1 Confidential Information?

This project does not contain any confidential or protected information.

2 IP to Protect

The team does not have any project-specific intellectual property that requires protection through an IP agreement. The DAIIM source code will be released as open-source software under the license described in the copyright license section below.

3 Copyright License

This project, including all source code and documentation, will be under the MIT license. The [license](#) can be found in the project repository.

4 Team Meeting Plan

Table 2: Team and Supervisor Meeting Schedule

Meeting	Location	Frequency	Time
Team meeting	McMaster campus	Weekly	Wednesday 1:30 pm
Supervisor meeting	Microsoft Teams	Biweekly	TBD

All meetings will be recorded in a Github issue outlining the date, attendance and agenda. Each meeting will have a chair who is responsible for creating the agenda and sharing it with others prior to the meeting. If possible the agenda should be shared 12hrs in advance to give team members a chance to review and propose different topics. For each meeting, the chair will ask for a volunteer from the team to record meeting minutes. If no one volunteers, the note-taking role will rotate among team members. Meeting minutes and decisions made during the meeting will be recorded as a comment on the meeting issue. Ad

hoc meetings may also be scheduled as necessary for urgent decisions, issue resolution, or preparation for a deliverable.

5 Team Communication Plan

The team will use a Microsoft Teams group chat for regular communication. The team also has a shared Word document for collaborative editing of documents and planning during meetings. Github issues will be used to track each meeting and any associated agendas/minutes.

Communication with the supervisor will be done through email and Microsoft Teams. Every email with the supervisor will be sent to all team members. The team will meet regularly with the supervisor, subject to the supervisor's availability.

Github issues will be the primary method of communication for development tasks. Each issue will represent a development task and will be assigned to a team member; the team member will be responsible for updating the issue with their progress. The issues will be associated with a milestone representing the deliverable, and will be placed appropriately in the Kanban project according to its status (e.g., To Do, In Progress, Done). Feedback is expected from at least one other team member for each issue, through either a comment on the Github issue or reviewed in a meeting or group chat.

6 Team Member Roles

- **Agenda creator:** The agenda creator is responsible for creating the agenda for a meeting, and posting it to the meeting issue. The agenda creator will use the template for team meetings located in the repo, containing the agenda and an attendance tracker.
- **Notetaker:** The notetaker is responsible for taking notes during a meeting, and posting the meeting minutes to the meeting issue. The meeting minutes will be posted as a comment under the meeting issue. Multiple people may contribute to the notetaker role during a meeting, especially if there are multiple topics / individual updates from each person.
- **Pull request reviewer:** The pull request reviewer role is a role that is assigned to a team member for each pull request. The pull request reviewer is responsible for reviewing the code changes in the pull request, providing feedback, and approving or requesting changes to the pull request. Every team member is expected to review pull requests, but the pull request reviewer is the primary reviewer for that pull request. They are in charge of making the final decision on approving the request.

- **Developer:** The developer role is a role assigned to a team member for each development task. The developer is responsible for implementing the task, and updating the associated issue with their progress. Every team member is expected to take on this role and contribute to development tasks as necessary, but the developer is the primary person responsible for completing a given task.

7 Workflow Plan

The following workflow plan can be used as a contributor's guide to the development process.

Issues and deliverables management: Each deliverable (and sub-deliverable if needed) will have a milestone associated with it. Each milestone will have a set of issues associated with it, representing the development tasks needed to complete the deliverable. These issues will be in the project Kanban board, and will be moved through the columns (To Do, In Progress, Done) as they are worked on.

Pull requests and code review: All code changes will be made in a new branch, and a pull request will be created once development is done for a particular task. See the branches and commit sections for naming convention. Typically, a branch will be created for each issue, but a branch may close several issues if they are related and can be completed together. Pull requests will be reviewed by at least one other team member, and the pull request reviewer role will be responsible for approving or requesting changes to the pull request.

7.1 Branches & Commits

Branch naming convention: Each branch will be named according to the following convention: `<branch-type>/<branch-name>`. Like mentioned, development will be done in a new branch, and a pull request will be created once development is done for a particular task. The naming convention for branches will be kebab-case and will be based on the branch content as well as the branch type. For example, a branch for the issue "Implement user login" would be named "feature/implement-user-login". The current types of branches are feature, docs, and bugfix, but more types may be added as needed.

Commit naming convention: Each commit will be named according to the following convention: `<commit-type>: <commit-message>`. The naming convention for commits will be lowercase and will be based on the commit content as well as the commit type. For example, a commit for the issue "Implement user login" would be named "feat: implement user login". The current types of commits are feat, fix, docs, style, refactor, and test, but more types may be added as needed.

7.2 Merging strategy

Developers will merge their changes into the main branch through pull requests. A pull request must be approved by at least one other team member before it can be merged into the main branch. Individual commits will be preserved in the pull request, and the pull request reviewer will be responsible for approving or requesting changes to the pull request.

8 Project Decomposition and Scheduling

The project is organized around key milestones and overarching stages (refer to Table 2), progressing from initial project planning and requirements analysis through design, implementation, verification, and final demonstration. Each stage contains defined deliverables and deadlines to provide the team with a clear progression throughout the project. The schedule will remain flexible where necessary, allowing the team to adjust tasks and responsibilities based on project needs, progress, and individual strengths.

Stage	Milestone	Deadline
Planning	Project selection	W2
Planning	Problem Statement + PoC + Development Plan + Team Charter	W4
Requirements	SRS + Hazard Analysis	W6
Verification	V&V Plan	W8
Design	Design Document	W10
Design	PoC Demo	W11
Implementation	Design Document Rev 0	W16
Implementation	Module Implementation	W16
Implementation	Rev 0 Demo	W18
Finalization	Final Demo + Documentation	W21
Finalization	V&V Report + CD	W22
Finalization	Final Demos	W23
Expo	Expo Materials	W25
Expo	Capstone EXPO	W26

Table 3: Project Milestones and Schedule

8.1 GitHub Project Usage

The project board will follow a Kanban-style workflow with columns for Backlog, Ready, In Progress, In Review, and Done. The project will be organized into milestones representing the overarching deliverables required throughout each stage of the project. For example, the initial stage contains milestones for the Project Statement and Development Plan, with individual tasks associated with each milestone. Each task will be represented as a card on the board and

assigned to the appropriate team member. Cards will move through the Kanban columns as work progresses, allowing the team to track the status of individual tasks and overall milestone completion.

9 Proof of Concept Demonstration Plan

9.1 Project Risks

One main risk is designing an aggregation algorithm that combines predictions from multiple inference providers into an accurate final result. The algorithm must handle disagreements between providers and reduce the influence of dishonest or low-quality predictions. This is difficult because, for a given task, the network may not have a source of *ground truth* against which to compare the predictions. There is also a cold-start problem where initially there may not be enough trusted providers in the network to correctly evaluate one another.

Another risk is designing the incentive system and token economics so that the network is sufficiently decentralized and is resistant to attacks. The system must be fair and accessible for participants so that they are encouraged to take part in the network. The system must also be economically sustainable and ensure that the distribution of rewards does not create incentives to exploit the system or reduce the quality of the aggregated results.

Collusion attacks pose a significant risk to the system. A collusion attack occurs when multiple participants, such as inference providers, coordinate their actions to manipulate the aggregation algorithm, reputations, and reward distribution. For example, providers could submit similar inaccurate predictions to make their results appear correct and receive rewards. This is difficult to detect when there is no ground truth and when the colluding providers make up a large portion of the network.

9.2 POC Demonstration Plan

The proof of concept plan is to create a simulation of the DAIIM network that is completely off chain. The simulation will execute a sequence of rounds in which an inference task for a regression problem is executed by the network. Various different types of actors will participate in the network:

1. Honest: attempt to maximize rewards by providing accurate predictions consistently (with some noise)
2. Learners: gradually improve their predictions over time
3. Malicious: intentionally provide bad predictions
4. Collusion: A group of providers working together to skew the network

The simulation will start with a simple ML task (eg. predict y , $y = mx + b$), and each round some input data will be randomly generated. The different actors will be given some algorithm (good or bad algorithm depending on the actor) to determine their predictions given the input. The actors will submit their predictions, and the aggregation algorithm will determine the aggregated

result. Actor's reputations will be updated and rewards will be distributed after each round.

Another layer of this simulation is to economically model each actor. We will assign providers a cost for running their model, transaction fees for submitting predictions on-chain, and economic rewards or penalties for participating. These values will be calibrated relative to the costs and payments that we expect a provider to realistically encounter when performing an inference task. This will ensure that the simulation reflects realistic participation decisions rather than relying on arbitrary values. We will also vary these values to measure how sensitive participation and network performance are to changes in costs, fees, rewards, and penalties.

We will test various combinations of actors to evaluate the limits of the system. These scenarios will vary the proportions of different actors, as well as the number of providers in the network. These results will allow us to identify conditions under which the system remains reliable and economically viable, as well as conditions under which malicious or colluding providers can significantly influence the network.

9.3 Success Criteria and Failure Response

This proof of concept will be successful and show that the stated risk can be overcome if: - The system provides accurate aggregated results that improve over time even with some limited number of malicious actors - It is more profitable to act honestly and attempt to provide good predictions than to be malicious - The network can be cold-started with the assumption that initially there are a majority of honest providers

If the proof of concept is not able to demonstrate that these risks can be addressed then the project's assumptions and scope may need to be changed. Before proceeding to full implementation, the team will review the findings with the supervisor and decide whether to revise the design, broaden the threat conditions, or reconsider the project's feasibility.

10 Expected Technology

This section describes the technologies that are expected to support the implementation of DAIIM. Note that these choices are preliminary and subject to change later in the project.

10.1 Core Implementation Stack

DAIIM will use Solana as the blockchain network for recording transactions and executing on-chain logic. During development, the team will use Solana Devnet, a separate network for testing programs and transactions without affecting Solana Mainnet. We will request free test SOL from a Devnet faucet to pay for transaction fees. Since Devnet tokens have no real-world value, the

team can develop and test DAIIM without using real money before deploying to Mainnet.

Solana programs, or smart contracts, will be written in Rust. The conventional off-chain backend will use Node.js with TypeScript and PostgreSQL for data storage. The user-facing interface will be developed with React.js and Vite. Python may also be used to write simple machine learning models for integration testing purposes.

10.2 Libraries and External Components

The expected external libraries and components are:

- *[Anchor]*: A development framework for building, deploying and testing Solana programs in Rust.
- *[Metamask]*: Used for accessing crypto wallets & signing transactions.
- *[Metaplex]*: Protocol and SDK for creating NFTs.
- *[@solana/web3.js]*: Javascript library for interacting with the Solana blockchain.
- *[@solana/spl-token]*: JavaScript library for interaction with the SPL token program on Solana.
- *[Vitest]*: Testing framework for the React frontend and other TypeScript code.
- *[Solana Devnet JSON-RPC]*: Public RPC access for testing blockchain transactions and programs.

10.3 Version Control and CI/CD

Git and Github will be used for version control and the project repository. GitHub Projects will organize issues and track their progress. LaTeX will be used for the project's technical documentation.

GitHub Actions will compile the LaTeX documentation automatically when relevant changes are pushed or proposed. Later, the CI pipeline will be extended to build and run tests for the solana program (using Anchor) and React frontend.

11 Use of AI and LLMs

12 Coding Standard

The team will adhere to the coding standards outlined in Table 4.

Category	Details
React Frontend	React code will follow the official React documentation and established JavaScript/TypeScript conventions. Code will be formatted using Prettier and linted using ESLint .
Rust Backend	Rust code will follow the official Rust API Guidelines and community style conventions, formatted using rustfmt and linted with clippy .
Rust Documentation	Rust functions, structs, and modules will use rustdoc doc comments (<code>///</code> and <code>//!</code>) following the conventions in the official Rust documentation guide .
Docker	Dockerfiles and container images will follow Docker's official best practices for maintainability, efficiency, and security.
Git	Git will be used for version control, with feature branches used to isolate development work.
Code Review	Changes will be submitted through GitHub pull requests and reviewed by another team member before being merged into the main branch. See the GitHub pull request review guidelines .

Table 4: Coding Standards

13 Appendix — Generative AI Use Disclosure

Below is an overview of how AI tools were used in the making of this document.

13.1 AI Tools Used

- Claude Sonnet 5.
- GPT-5.6 Luna.
- Gemini 1.5.

13.2 How and Where Used

Some members used AI to aid in explaining certain machine learning topics and applications, as there were varying levels of experience in the subject matter.

Additionally, AI was used to facilitate some brainstorming. This occurred in parts of the Team Charter as supplementary ideas for elements of a code of conduct and ways to reward members for positive contributions. It was also used in the Expected Technology section to generate a list of potential libraries that could be used in the project.

AI tools were also used throughout this document to check for grammar and spelling errors, but were not used to write whole sentences or significantly

rewrite text that had been written by members. Occasionally, AI tools were also used to learn LaTeX syntax (how to insert a hyperlink, for example).

13.3 Verification of AI-Generated Content

The use of AI to explain complicated subject matter was used in conjunction with reading reliable sources, such as the Solana Handbook.

The instances in which AI was used for brainstorming related to the team charter were purely for basic idea generation and was up to the discretion and opinion of the group to deem them valuable to the project. The list of potential libraries that was generated was used as a means to build awareness to existing available tools. The decision to include specific libraries made based on further research into the official documentation of the libraries, not the AI response.

When AI tools were used to check grammar and spelling, team members still critically reviewed the document in its entirety to ensure the document was still understandable and double check the grammar and spelling.

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. Why is it important to create a development plan prior to starting the project?

Creating a development plan prior to starting the project helps to encourage thoughtful communication between team members and facilitates helpful discussions regarding expectations for the individual and for the group.

It helped to ensure that we were all on the same page and helped us to think about the future of the project. Outlining the Proof of Concept plan helped us to confront what is reasonable within the scope of the project, and we were able to identify possible pain points. For example, we realized that the UI will most likely be a large part of the project, but is not necessary to be super involved. In contrast, we found that we are still shaky on how we will implement the reputation system. Thus, this helped us to narrow our focus and identify the expected pain points of the initial development of the system.

Outlining roles and assigning actual metrics to the expectations really helped with administrative organization and getting a better understanding of how we want to work both as a group and individually. For example, conflict resolution was not something we had considered. However, we have now developed a system that will hopefully help in catching issues early, hold everyone accountable, and address the issues in a respectful, agreed-upon manner. It also helped us to think about the necessity of having a plan for consistent contributions and to have deadlines throughout the project to keep the team on track and avoid major crams.

While we understand that the information laid out in this development plan may not end up going completely as expected, creating this document led to fruitful discussions and agreements, and helped us to get a better understanding of the project as a whole.

2. In your opinion, what are the advantages and disadvantages of using CI/CD?

CI/CD can be extremely helpful, especially in group projects, when executed correctly. It can help to outline certain standards. For example, having a set of rules that must be adhered to in order for a pull request to be accepted.

It can also help to prevent bugs from unknowingly arising when changes are made to existing code and disrupting code that was previously written off as meeting requirements and passing test cases. This is especially useful in a project with many collaborators and on complex projects, as small bugs introduced can be hard to catch and to pinpoint when they were introduced.

However, CI/CD can be complicated to implement and can add to project overhead.

As a group, we have decided to implement guidelines that must be followed when merging a branch into Main (no conflicts, at least one reviewer), the existing automatic PDF generator (with potential modifications as the project progresses), and the addition of unit testing where deemed necessary. We may choose to add more CI/CD elements to the project in the future, at the discretion of the group.

3. What disagreements did your group have in this deliverable, if any, and how did you resolve them?

There were no disagreements in this deliverable. Everyone was respectful in sharing their ideas and listening to others. Any misunderstandings that were voiced were met with non-judgmental education, and agreements were decided from there. For example, there was uncertainty regarding the feasibility and expectations for the Proof of Concept. This led to educational discussions, constructive criticism, and educational moments.

The team navigated these discussions with respect and open-mindedness, and we believe this bodes well for the future of the project, conflict resolution, and the team dynamic.

Appendix — Team Charter

External Goals

For this Capstone, our team hopes to deliver a project that is both challenging and enjoyable to create. We aim to build something beyond simply fulfilling the requirements of a school project—something that we can be proud of and potentially carry forward into our professional careers. By focusing on producing a meaningful, well-developed, and high-quality project, we believe that a strong performance in the Capstone course will naturally follow.

Attendance

This section summarizes the ground rules and expectations regarding team member attendance.

Expectations

All team members are expected to attend weekly team meetings and meetings with our supervisor. Attendance at every Capstone course lecture is not required, provided that members remain up to date with the course material through either attending the lecture in person or reviewing the material independently.

Attendance at all required team and supervisor meetings will be tracked through GitHub Issues. All members are expected to maintain a 95% attendance rate throughout the project. When a meeting is held in person, team members should

make their best effort to attend in person. If a member is unable to attend in person, they should notify the team at least 5 hours in advance and attend virtually instead. If a member is unable to attend the meeting altogether, they must notify the team as soon as possible, preferably when they become aware of the conflict and at least 24 hours before the meeting when reasonably possible. Different expectations may apply depending on the circumstances.

Acceptable Excuse

Team members are expected to communicate with the rest of the team if they are unable to attend a meeting or meet a project deadline. Each member may miss up to three team meetings per term with prior notice. Acceptable reasons may include illness, family emergencies, religious observances, or previously arranged commitments that cannot reasonably be rescheduled.

If a member exceeds three absences in a term, the team will discuss the member's participation, expectations, and responsibilities. If the issue cannot be resolved

within the team, the team may involve the course professor or supervisor.

Team members are expected to meet all assigned deadlines. Exceptions will generally only be made in the event of an emergency, as outlined in the following section.

In Case of Emergency

If a team member is unable to meet a deadline or complete their assigned work due to an emergency, they must communicate this to the team as soon as they become aware that the work cannot be completed on time. The team will then

discuss appropriate accommodations and, if necessary, redistribute the affected work to ensure that project deadlines can still be met. If another team member must take on the incomplete work, the original team member will be expected to take on additional responsibilities during a subsequent phase, where appropriate, to help maintain a fair distribution of workload.

[borrows from [University of Portland Team Charter](#) —SS]

Accountability and Teamwork

Quality

To reach our goal of using this capstone experience to build something that we as a team are proud of, all members agree to make contributions that meet or exceed expectations. Level 4 of the rubric of deliverables will be referenced and used as a guideline for minimum expected requirements.

Team members are also expected to come to meetings with their assigned tasks completed and prepared to share their progress. Members are expected to actively contribute to team discussion and provide constructive criticism to help augment the quality of the project.

During code reviews, reviewers are expected to provide at least one point of constructive criticism to encourage consistent improvement.

Attitude

All team members will contribute to creating a welcoming and non-judgmental environment in which members can feel comfortable openly sharing ideas. All ideas will be considered in their entirety and seen as valuable contributions to the development process, whether they are accepted or viewed as opportunities to provide constructive criticism and identify miscommunications or misunderstandings.

Members are expected to communicate respectfully and refrain from interrupting or speaking over others. Everyone should be actively listening during meetings and provide thoughtful insight to team discussions.

All members will communicate respectfully and professionally. Contributions will be acknowledged and openly appreciated. Cooperation is expected

between team members and can be showcased through a willingness to learn, to compromise, to help fellow members when reasonably possible, and to give useful feedback during code reviews.

All disagreements are expected to be handled professionally and constructively. Should conflict arise, members will take a reasonable break until they feel that they can address it in a productive and professional way. If conflict arises that cannot be resolved by the team, a meeting with the TA and professor will be requested, and the entire team is expected to attend, even if certain members are not directly involved.

Stay on Track

To ensure deadlines are consistently met and the project stays on track, a [GitHub kanban project](#) will be used and maintained to stay up to date on all issues and milestones/deliverables. All issues will be labelled with the type, the associated milestone, the assignee(s), and the target deadline for the issue to be completed and ready for review.

Deadlines will be assigned to each issue to encourage incremental, iterative, and consistent progress, rather than big pushes before externally enforced deliverable deadlines. The schedule and availability of members will be taken into consideration when assigning tasks and deadlines, to ensure that tasks and timelines are manageable and do not cause unnecessary stress. Tasks will also be assigned with a goal of sharing an even workload amongst all team members (1/4 of the expected workload). Team members will be free to voice their concerns if the workload does not seem even, and the task assignments will be reviewed and reassessed.

Members should update the status of their issues regularly as the status changes. When an issue is ready for review, the assignee will create a pull request and assign a primary reviewer. The primary reviewer must respond and give feedback within 48 hours. Team member contributions will be assessed based on Git activity such as commits, pull requests, and issues completed to show consistent contributions that meet quality standards, as well as meeting the planned deadlines.

At weekly team meetings, members will be expected to share the status of what they have been working on that week, as well as action items for what they intend to work on or complete before the next meeting. Each member should have an attendance record of > 95%. If consistent lateness becomes an issue, it will be addressed by the team.

In the case of the anticipation of a missed deadline, members should raise concerns about meeting deadlines at least 48 hours before the deadline to ensure the team can adjust and plan for the work to still be completed. In the case of a member needing to offload their assigned tasks onto other members to meet the deadline, it will be handled as a missed deadline and will be handled according to what has been outlined above in **Attendance**.

Members who are not meeting expectations will be respectfully confronted at the following team meeting. They will have the opportunity to explain their

situation and must come up with a detailed plan describing how they will get back on track. This plan will be reviewed and approved by the other team members, and their progress will be reviewed at each subsequent team meeting until the member consistently meets expectations. If at the following meetings it becomes clear that little to no effort is being made to get back on track, the issues will be brought to the attention of the TA in an effort to seek external support in conflict resolution.

Members who consistently meet or exceed expectations will be recognized by the team for their contributions. This may include informal recognition during team meetings, opportunities to take on preferred tasks, or small team rewards after completing major milestones. If the entire team reaches a milestone ahead of schedule while maintaining the expected quality of work, the team may use the additional time for a team social activity or other agreed-upon reward.

Team Building

Team building activities will be planned and hosted biweekly. Potential events may include, but are not limited to, team lunches and trivia nights. The events will be planned for a time that works for all members' schedules and will be mindfully inclusive of the needs and interests of all team members.

Decision Making

Team decisions will occur through consensus, first allowing all team members to voice their opinions and concerns. The team will collectively come to an agreement, with the success of the team and project as a whole as a priority.

In the case of a disagreement, each member will be treated with respect and given the opportunity to express their opinions honestly and respectfully. Efforts will be made to come to a healthy compromise.

If time constraints allow, any disagreement that cannot be resolved within the meeting will be revisited at a subsequent meeting. All members of the team will take time between meetings to view the situation from a different perspective to encourage empathy in the decision-making process and academic research, if necessary.